

SelahOS

Native Apps

Installation & Usage Guide

SelahBrowser

WebKit-powered browser with ad blocking

Muse

Claude AI assistant * Meta+M

Omolara

ChatGPT assistant * Meta+O

SELAH-30 * Selah Technologies LLC * selahos.com

Build: C++17 * Qt6 * WebKit2GTK * SQLite * XCB

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1 * Prerequisites & Build Machine

All three apps are built with CMake + Ninja on the SelahOS build machine (Arch Linux, linux-zen kernel). They target the SelahOS live environment and are packaged directly into the ISO.

Required packages (installed by install-deps.sh)

All apps:

- cmake >= 3.16
- ninja
- qt6-base, qt6-quick, qt6-quickcontrols2, qt6-declarative
- sqlite (Qt SQL driver)
- xcb-util-keysyms, libxcb (global hotkeys)
- pkgconf

SelahBrowser only:

- webkit2gtk (webkit2gtk-4.0, GTK3-based)

Build machine setup

```
# On SelahOS (Arch Linux)
sudo pacman -Syu
cd ~/selahbrowser && bash install-deps.sh
cd ~/muse          && bash install-deps.sh
cd ~/omolara       && bash install-deps.sh
```

These scripts call pacman --needed so re-running is safe.

2 * SelahBrowser

SelahBrowser is a Qt6-native web browser using WebKit2GTK for rendering. The Qt chrome (tab bar, address bar) sits above a WebKitView widget that renders pages via an offscreen GTK surface.

2.1 Installation

```
git clone https://github.com/SelahOS/selahbrowser
cd selahbrowser
bash install-deps.sh          # installs webkit2gtk + qt6 deps
cmake -B build -G Ninja \
    -DCMAKE_BUILD_TYPE=Release
ninja -C build
sudo ninja -C build install   # installs to /usr/bin/selahbrowser
```

2.2 Configuration

SelahBrowser stores its settings in `~/.config/selah/browser/`. On first launch the directory is created automatically.

- Settings file: `~/.config/selah/browser/settings.ini`
- Ad-block rules: `~/.config/selah/browser/adblock.txt` (EasyList format)
- Window geometry persists automatically across sessions.

```
# Example settings.ini
[Browser]
home_page=https://search.brave.com
default_zoom=1.0
power_mode=performance    # or: battery
```

2.3 Using SelahBrowser

New tab	Click the + button in the tab bar, or Ctrl+T
Close tab	Hover over tab -> click x, or Ctrl+W
Navigate	Click the address bar, type URL or search query, press Enter
Search	Type a plain query in the address bar -- routes to Brave Search
Back / Forward	<- -> buttons in the toolbar
Reload / Stop	[r] / x button (changes during page load)
Power mode	Click the [!]Perf or [bat]Save pill to toggle
Battery mode	Disables WebGL, HW acceleration, forces click-to-play media
Tab auto-suspend	Tabs idle for 5+ minutes are suspended in Battery mode
Ad blocking	Enabled by default -- 25+ tracker domains + path patterns
Find on page	Programmatic: <code>browser.findText("query")</code> via QML API
Zoom	Set in <code>settings.ini</code> ; runtime: <code>browser.setZoom(factor)</code>

SelahBrowser uses webkit2gtk-4.0 (GTK3). Input forwarding and rendering work on X11/XWayland. Pure Wayland (no XWayland) requires a compositor subsurface integration not yet included in this release.

3 * Muse (Claude AI Assistant)

Muse is the SelahOS-native Claude interface. It lives in the system tray, springs open with Meta+M, and streams responses from the Anthropic API. Conversation history is stored locally in SQLite.

3.1 Installation

```
git clone https://github.com/SelahOS/muse
cd muse
bash install-deps.sh
cmake -B build -G Ninja \
      -DCMAKE_BUILD_TYPE=Release
ninja -C build
sudo ninja -C build install # installs to /usr/bin/muse

# Auto-start with the session (optional)
cp resources/muse.desktop \
  ~/.config/autostart/muse.desktop
```

3.2 API Key Setup

Muse reads your Anthropic API key from two sources, checked in order:

- Environment variable: `ANTHROPIC_API_KEY`
- Settings file: `~/.config/selah/muse/settings.ini`

```
# Option A -- environment variable (recommended)
# Add to ~/.zshrc or ~/.bash_profile:
export ANTHROPIC_API_KEY="sk-ant-api03-..."

# Option B -- settings file
mkdir -p ~/.config/selah/muse
cat > ~/.config/selah/muse/settings.ini << 'EOF'
[Muse]
api_key=sk-ant-api03-...
model=claude-sonnet-4-6
EOF
```

The API key is never written back to disk by Muse. Option A (env var) keeps it out of any file on disk.

3.3 Using Muse

Launch

muse starts minimised to tray. No window appears until triggered.

Show / hide	Press Meta+M (Super+M) from anywhere on the desktop.
Show via tray	Left-click or double-click the Muse tray icon.
Send message	Type in the input box at the bottom. Press Enter to send.
New line	Shift+Enter inserts a newline without sending.
Cancel stream	Click the [stop] stop button while the AI is responding.
New conversation	Click the + button in the top-right of the header bar.
Switch history	Click any conversation in the left sidebar.
Change model	Click the model pill (e.g. 'Sonnet 4.6') to cycle to Haiku.
Close window	Press Escape or close normally -- app stays in tray.
Quit	Right-click tray icon -> Quit.

Available models

```

claude-sonnet-4-6          # default -- best quality
claude-haiku-4-5-20251001 # faster, cheaper

```

4 * Omolara (ChatGPT Assistant)

Omolara is the SelahOS-native OpenAI interface. Identical UX to Muse -- tray app, streaming responses, local SQLite history -- but powered by GPT-4o.

4.1 Installation

```
git clone https://github.com/SelahOS/omolara
cd omolara
bash install-deps.sh
cmake -B build -G Ninja \
      -DCMAKE_BUILD_TYPE=Release
ninja -C build
sudo ninja -C build install # installs to /usr/bin/omolara

# Auto-start with the session (optional)
cp resources/omolara.desktop \
  ~/.config/autostart/omolara.desktop
```

4.2 API Key Setup

Omolara reads your OpenAI API key from:

- Environment variable: OPENAI_API_KEY
- Settings file: ~/.config/selah/omolara/settings.ini

```
# Option A -- environment variable (recommended)
export OPENAI_API_KEY="sk-proj-..."

# Option B -- settings file
mkdir -p ~/.config/selah/omolara
cat > ~/.config/selah/omolara/settings.ini << 'EOF'
[Omolara]
api_key=sk-proj-...
model=gpt-4o
EOF
```

4.3 Using Omolara

Launch	omolara starts minimised to tray.
Show / hide	Press Meta+O (Super+O) from anywhere on the desktop.
Show via tray	Left-click or double-click the Omolara tray icon.
Send message	Type in the input box. Press Enter to send.
New line	Shift+Enter inserts a newline without sending.
Cancel stream	Click the [stop] stop button while responding.

New conversation

Click + in the header.

Switch history

Click any conversation in the left sidebar.

Change model

Click the model pill to cycle between GPT-4o and GPT-4o mini.

Close window

Escape or close normally -- stays in tray.

Quit

Right-click tray icon -> Quit.

Available models

```
gpt-4o          # default -- best quality
gpt-4o-mini     # faster, cheaper
```


5 * Shared Configuration

All three apps store their data under ~/.config/selah/. Each app has its own subdirectory.

```
~/.config/selah/  
??? browser/  
?   ??? settings.ini      # home page, zoom, power mode  
?   ??? adblock.txt       # custom EasyList rules (optional)  
??? muse/  
?   ??? settings.ini      # model preference (api_key optional)  
?   ??? history.db        # SQLite conversation history  
??? omolara/  
    ??? settings.ini      # model preference (api_key optional)  
    ??? history.db        # SQLite conversation history
```

Resetting an app

```
# Reset Muse (clears all history and settings)  
rm -rf ~/.config/selah/muse  
  
# Reset SelahBrowser (clears settings only, keeps adblock rules)  
rm ~/.config/selah/browser/settings.ini
```

history.db files are plain SQLite -- you can open them with any SQLite browser (e.g. DB Browser for SQLite) to export, backup, or inspect conversations.

6 * Building from Source

Each app is a standalone CMake project. No Makefile wrappers needed.

General build flow

```
cd <app-directory>  
cmake -B build -G Ninja \  
      -DCMAKE_BUILD_TYPE=Release \  
      -DCMAKE_INSTALL_PREFIX=/usr  
ninja -C build  
sudo ninja -C build install
```

Debug build

```
cmake -B build-debug -G Ninja \
      -DCMAKE_BUILD_TYPE=Debug
ninja -C build-debug
./build-debug/<appname> # run directly without installing
```

Build outputs

- Binary: /usr/bin/<appname>
- .desktop: /usr/share/applications/<appname>.desktop
- Icon: /usr/share/icons/hicolor/scalable/apps/<appname>.svg

Qt6 moc and QML compilation happen automatically via CMake's qt_add_executable / qt_add_qml_module -- no manual qmake steps needed.

7 * Adding Apps to the ISO

To ship all three apps in the SelahOS ISO, add the following to the airootfs build process.

Build step in customize_airootfs.sh

```
# Build and install all three apps into the ISO rootfs
for app in selahbrowser muse omolara; do
  cd /tmp/$app
  cmake -B build -G Ninja -DCMAKE_BUILD_TYPE=Release \
        -DCMAKE_INSTALL_PREFIX=/usr
  ninja -C build
  DESTDIR=/mnt/airootfs ninja -C build install
done
```

Package list additions (packages.x86_64)

```
webkit2gtk
qt6-base
qt6-quick
qt6-quickcontrols2
qt6-declarative
xcb-util-keysyms
sqlite
```

Autostart entries for liveuser

```
# In customize_airootfs.sh, after creating liveuser:  
install -d /etc/skel/.config/autostart  
install -m644 /usr/share/applications/muse.desktop \  
    /etc/skel/.config/autostart/  
install -m644 /usr/share/applications/omolara.desktop \  
    /etc/skel/.config/autostart/
```

SelahBrowser should NOT be set to autostart. Set it as the default browser in KDE System Settings -> Applications -> Web Browser.

8 * Troubleshooting

SelahBrowser shows a blank black screen

The WebKitView offscreen surface hasn't rendered yet, or webkit2gtk failed to initialize. Check that webkit2gtk-4.0 (not 4.1) is installed:

```
pacman -Q webkit2gtk
```

Run selahbrowser from a terminal to see GLib/WebKit error output.

SelahBrowser input (keyboard/mouse) not working

Input forwarding uses GTK3 GdkEvent injection which requires X11/XWayland. If you're on a pure Wayland session with no XWayland, set:

```
DISPLAY=:0 selahbrowser
```

or enable XWayland in your compositor config.

Meta+M / Meta+O hotkey not triggering

XCB key grabs require X11. On Wayland, ensure XWayland is running. You can verify:

```
echo $DISPLAY # should print :0 or similar
```

If the key is grabbed by another app, the grab will silently fail.

Muse / Omolara: 'ANTHROPIC_API_KEY is not set'

The API key is missing. Set it in your shell profile:

```
echo 'export ANTHROPIC_API_KEY=sk-ant-...' >> ~/.zshrc
source ~/.zshrc
```

Then relaunch the app.

Muse / Omolara: network error / 401 Unauthorized

Your API key is invalid or expired. Generate a new one at:

platform.anthropic.com (Muse)

platform.openai.com (Omolara)

Muse / Omolara: no tray icon visible

KDE Plasma sometimes delays tray icon registration. Right-click the system tray -> System Tray Settings -> Extra Items and make sure Muse/Omolara are enabled.

CMake: 'webkit2gtk-4.0 not found'

Install the package:

```
sudo pacman -S webkit2gtk
```

Note: webkit2gtk and webkit2gtk-4.1 are different packages.

Qt6 modules not found during cmake

Install the full Qt6 set:

```
sudo pacman -S qt6-base qt6-quick qt6-quickcontrols2 qt6-declarative
```

Then re-run cmake (delete the build/ directory first).

history.db locked / can't open

Another instance of the same app is running. Each app uses a unique database connection name, but two instances sharing the same db file will conflict. Kill the other instance:

pkill muse

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